Zero-Trust P2P: A Cryptographic and Reputation-Based Framework for Secure Decentralized File Sharing

Abstract

While popular Peer-to-Peer (P2P) systems like BitTorrent, uTorrent, and IPFS enable high-speed direct file transfers without centralized servers, they fundamentally operate on blind trust. Because any anonymous participant can join the network, attackers can easily inject corrupted, fake, or malware-laden data chunks (content poisoning), forcing users to waste substantial bandwidth and time on failed downloads. Furthermore, existing clients lack long-term memory, allowing malicious peers to simply reconnect under new addresses and repeat attacks without consequence. To solve these vulnerabilities, we propose Zero-Trust P2P, a secure decentralized file-sharing framework built on the principle of "never trust, always verify." Rather than waiting for an entire file to complete, our system employs chunk-level Merkle Tree cryptographic hashing to validate the authenticity of every incoming block the millisecond it arrives, discarding corrupted data before it touches the user's storage. In parallel, a dynamic peer reputation engine tracks historical node behavior, rewarding honest contributors while immediately penalizing and blacklisting peers that transmit corrupted blocks across the swarm. By combining instantaneous integrity checks with continuous reputation tracking, Zero-Trust P2P eliminates bandwidth waste, ensures 100% data integrity, and delivers a resilient, secure foundation for decentralized data sharing.

Software & Hardware Requirements

Operating System: Windows or Ubuntu Linux
Programming Language: Python 3.10+
Core Libraries:

- hashlib & cryptography (SHA-256 Merkle tree verification, digital signatures for peer IDs)
- socket & asyncio (peer-to-peer TCP/UDP connection management)
- sqlite3 / tinydb (local storage for peer reputation history and blacklist tables)

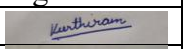
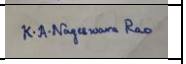
Testing & Interface:

- Streamlit (lightweight web dashboard to visualize connected peers, live trust scores, and security alerts)
- Docker (optional, for simulating multiple peer nodes on a single machine)

Processor: Intel Core i5 / AMD Ryzen 5 or higher (minimum 4 cores)
RAM: 8 GB (standard laptop configuration)
Storage: 20 GB free space (SSD recommended)
Network: Standard Wi-Fi / Ethernet connection with local loopback support

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